Regression ensemble and classifier: stage-specific inputs

Stages I–III	Inputs: $k_B T/t,\ J_A/t$ Two hidden layers: 64 + 64 ReLU units
Stage IV	Inputs: $k_B T/t,\ J_A/t,\ \mu/t$ Two hidden layers: 96 + 96 ReLU units
Both tasks	Five independent members per task; retrain each round Regression: MSE; classification: cross entropy
Outputs	Regression: $\Delta, q, \eta, I_c^+, I_c^-$ and dispersion Classification: phase probabilities

Acquisition is a scheduler only; exact BdG results supply the labels.